

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Fourth Semester of B. Tech. Examination (CL)****April - May 2018****CL 245 Building Construction Technology****Date: 02.05.2018, Wednesday****Time: 10:00 a.m. To 01:00 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q - 1 Do as directed:**

- i. Groove provided on the underside of the projecting elements such as sill cornices, copings etc. so that the rainwater can be discharged clear of the surface is called **[01]**
 - a) Cornice
 - b) Weathering
 - c) Throating
 - d) Tooothing
- ii. A staircase which turns direction 3 times with the upper flight crossing the bottom one is called _____ staircase. **[01]**
- iii. Give any one thumb rule to be followed for designing a comfortable staircase. **[01]**
- iv. Draw a figure of sloping roof building showing following elements **[02]**
 - a) Ridge
 - b) Eaves
- v. What type of pre-fabricated acoustic material should be used to provide sound insulation to large open rooms? **[01]**
- vi. Developments of fissures in plaster due to movements in background or surrounding structure is called _____. **[01]**

Q – 2.a Write a short note on Ashlar fine tooled masonry. **[04]****Q – 2.b** What are the essential features of English bond masonry? Explain with the help of a neat sketch of the elevation of a wall constructed in English bond. **[05]****OR****Q – 2.b** Explain the advantages of hollow concrete block masonry. **[05]****Q – 2.c** What are the requirements of a good stair? **[05]****OR****Q – 2.c** Plan an open newel half turn staircase for a commercial building in which the vertical distance between each floor is 3.6 m. The size of stair hall is limited to 4 m x 5 m. **[05]****Q – 3.a** What are the advantages and disadvantages of flat terraced roofing? **[04]**

Q – 3.b What methods of fire resistant construction should be adopted for fire escape elements of a building? [05]

OR

Q – 3.b Apart from providing insulating material on building elements, what other methods can be employed to provide thermal insulation in a building? Explain in detail. [05]

Q – 3.c What are the fundamental principles of green building design? Explain energy efficiency in detail. [05]

OR

Q – 3.c How do the regular buildings effect the environment? Explain the benefits of green buildings. [05]

SECTION – II

Q - 4 Do as directed: [07]

- i. The total area of window-openings should normally vary from _____ to _____ % of the floor area of the room, depending upon the climatic conditions.
- ii. The slab provided when length of room is greater than 1.5 times width of room is called _____ slab.
- iii. What kind of flooring should be provided in libraries, theatres, art galleries etc to make them noiseless?
 - a) Marble flooring b) Cement concrete flooring
 - c) Cork flooring d) Asphalt flooring
- iv. What kind of scaffolding should be used for stone masonry?
- v. Needle scaffolding is also known as _____.
- vi. For a swimming pool project which kind of damp proofing method will be adopted?
- vii. Hot bitumen as a damp proofing material can be applied with a minimum thickness of 3 mm. True or False?

Q – 5.a What are the various functions served by window? [02]

Q – 5.b i. In which conditions should the pile foundation be adopted? [02]
 ii. Explain isolated footing with a neat sketch. [05]

OR

Q – 5.b Design a strip footing for a brick wall 30 cm thick and 3.5 m high above ground level. [07]
 The wall carries a superimposed load of 120 KN per meter run. The soil has unit weight of 17 KN/m³, angle of repose of 30° and safe bearing capacity of 160 KN/m².

The footing may have lime concrete base, which has unit weight of 20 KN/m^3 and modulus of rupture equal to 160 KN/m^2 . Take the unit weight of masonry as 19.5 KN/m^3 .

Q – 5.c What points should be kept in mind while locating doors and windows in a building? **[05]**

OR

Q – 5.c Explain battened and ledged door with a neat sketch. **[05]**

Q – 6.a i. Explain the symptoms and reasons for the following defects in plastering. **[03]**

a) Blistering

b) Popping

c) Cracking and Crazeing

ii. Explain various causes of dampness in buildings. **[03]**

Q – 6.b Explain any eight factors affecting the choice of flooring materials. **[04]**

OR

Q – 6.b Describe the procedure of constructing cement concrete flooring. **[04]**

Q – 6.c What are the various types of scaffolding? Describe single scaffolding with a neat sketch. **[04]**

OR

Q – 6.c Enlist various methods for timbering in trenches and explain any one in detail. **[04]**
